

Dimension points

Part BLANK_ERSB06-07G_PAN_EXTRUDED. Solid of revolution about the Z axis. Source file BLANK_ERSB06-07G_PAN_EXTRUDED.step, written 2026-09-29T12:34:42 by build123d / Open CASCADE. Every model-space coordinate below is taken from that file. Units are millimetres.

How to read the points

The boundary is stored as one seam on the +X side of each circular edge, plus the bottom pole. P01 through P10 are those points, in boundary order from the bottom pole to the top face. A point with radius R is the +X sample of a full circle: centre (0, 0, Z), diameter 2R. That circle is the complete set of dimension points at the station. The spline-control register at the end lists every other 3D coordinate stored in the file.

Overall size

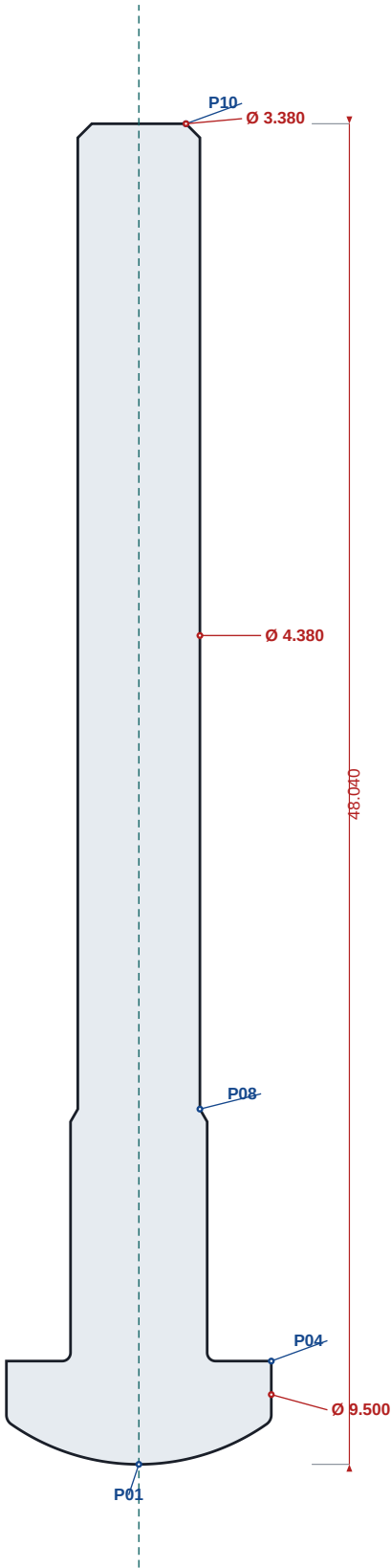
Dimension	Value (mm)	From	To
Overall length, along Z	48.040	P01 Z -3.700	P10 Z 44.340
Overall diameter	Ø 9.500	P03	P04
Top-face diameter	Ø 3.37962	P10	closed disk at Z 44.340
Datum	Z = 0	flange face	axis through (0, 0)

Z = 0 is the flat flange face. Positive Z runs from that face toward the tip. The base sphere hangs below the datum, so the bottom pole is at Z -3.700.

Profile drawing — full length

Section through the Z axis

true proportion · +X seam points

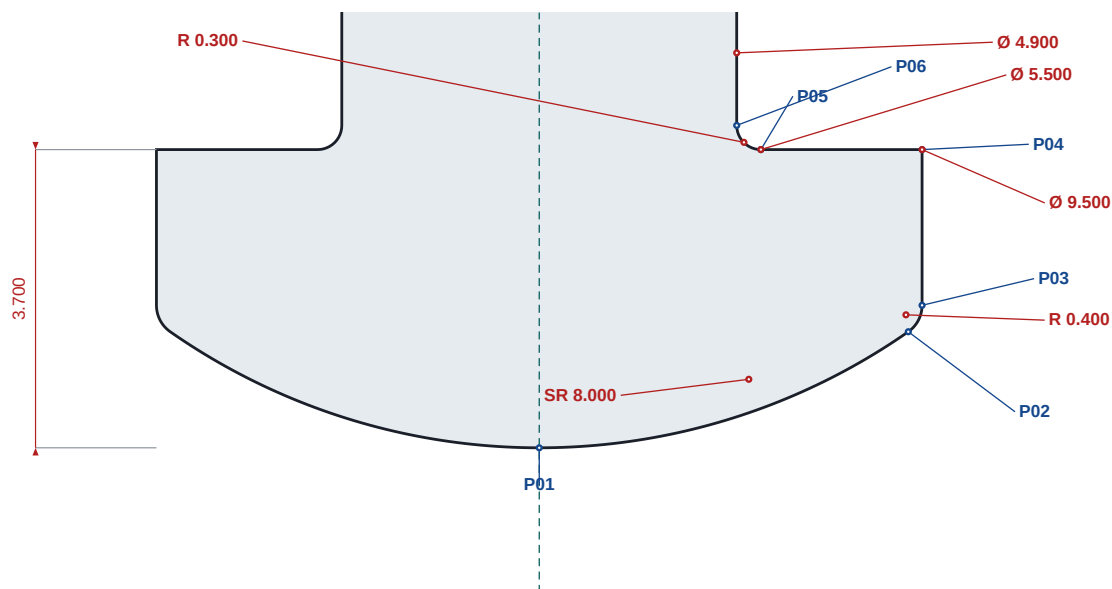


Base — points P01 to P06

Spherical base SR 8.000 centred at (0, 0, 4.300), blended by an external R0.400 arc into the Ø9.500 cylinder. The cylinder ends on the datum face. From P04 the face runs in to P05, then an external R0.300 round drops onto the Ø4.900 stem.

Base detail

Z -5.50 to 1.70



The R0.400 centre is (4.350, 0, Z of P03). It is tangent to the sphere at P02 and tangent to the Ø9.500 cylinder at P03. The R0.300 centre is (2.750, 0, 0.300): a quarter-circle from P05 to P06.

Stem taper — points P07 and P08

Straight $\varnothing 4.900$ stem ends at P07. A 30° semi-angle taper runs to the neck at Z 9.030 (P08). P07's station and the neck radius are the intersection of that taper with the stem and with Z 9.030; both are driven values and are quoted from the STEP literals.

30 degree taper

Z 8.05 to 9.75

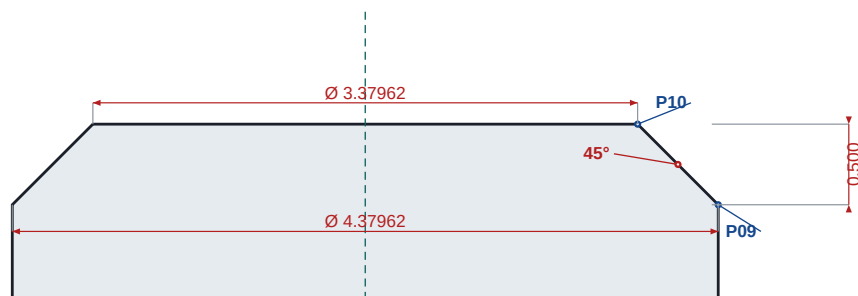


Tip chamfer — points P09 and P10

The neck is a cylinder at the P08 radius from Z 9.030 to Z 43.840. P09 to P10 is a 45° chamfer, 0.500 mm on the radius and 0.500 mm on Z. The top face is a closed disk at Z 44.340.

45 degree tip chamfer

Z 43.25 to 45.05



Seam dimension points P01–P10

Coordinates are the STEP vertex coordinates. Radius is the distance to the Z axis. Diameter is the full circle through that point. Values below 0.000001 mm are written as 0.

Point	Feature	X	Y	Z	Radius	Diameter
P01	Bottom pole	0.000	0.000	-3.700	0.000	—
P02	Sphere / R0.4 blend	4.578947	0.000	-2.259973	4.578947	Ø 9.157895
P03	Blend / outside cylinder	4.750	0.000	-1.931974	4.750	Ø 9.500
P04	Flange, outside	4.750	0.000	0.000	4.750	Ø 9.500
P05	Flange, inside	2.750	0.000	0.000	2.750	Ø 5.500
P06	Round / stem cylinder	2.450	0.000	0.300	2.450	Ø 4.900
P07	Stem / 30° taper	2.450	0.000	8.579338	2.450	Ø 4.900
P08	Taper / neck	2.18981	0.000	9.030	2.18981	Ø 4.37962
P09	Neck / 45° chamfer	2.18981	0.000	43.840	2.18981	Ø 4.37962
P10	Chamfer / top face	1.68981	0.000	44.340	1.68981	Ø 3.37962

Stations along the boundary

From	To	Δ radius	Δ Z	What it is
P01	P02	4.578947	1.440027	Sphere, from the pole to the blend tangency
P02	P03	0.171053	0.327999	R0.400 blend
P03	P04	0.000	1.931974	Ø9.500 cylinder, up to the datum face
P04	P05	-2.000	0.000	Flange face, radially inward
P05	P06	-0.300	0.300	R0.300 external round
P06	P07	0.000	8.279338	Ø4.900 stem cylinder
P07	P08	-0.26019	0.450662	30° taper
P08	P09	0.000	34.810	Neck cylinder
P09	P10	-0.500	0.500	45° chamfer, 0.500 × 0.500

Feature dimensions

Analytic surfaces and the circles that bound them. A semi-angle is the angle between the cone axis and the cone generator. Design values that land on a clean millimetre are marked exact; the remaining figures are the driven STEP values.

Feature	Size	Location	Status
Spherical base	SR 8.000	Centre (0, 0, 4.300)	Exact
Bottom pole P01	Z −3.700	(0, 0, −3.700)	Exact
Base blend	R 0.400	Centre (4.350, 0, Z of P03), tangent at P02 and P03	Exact radius
Blend tangency P03	Z -1.931974	X 4.750, Y 0	Driven
Outside cylinder	Ø 9.500	From Z -1.931974 to Z 0.000	Exact diameter
Flange face	Z 0.000	Annulus from Ø5.500 to Ø9.500	Exact
Flange radial width	2.000	P04 to P05	Exact
Stem round	R 0.300	Centre (2.750, 0, 0.300), quarter-circle P05 to P06	Exact
Stem cylinder	Ø 4.900	From Z 0.300 to Z 8.579338	Exact diameter
Stem length	8.279338	P06 to P07, parallel to Z	Driven
Taper	30° semi-angle	Ø4.900 at P07 to Ø4.37962 at Z 9.030	Exact angle
Taper end station	Z 9.030	P08	Exact station
Neck cylinder	Ø 4.37962	From Z 9.030 to Z 43.840	Driven diameter
Neck length	34.810	P08 to P09	Exact
Tip chamfer	0.500 × 45°	Radial drop 0.500 and axial rise 0.500	Exact
Top face	Z 44.340	Closed disk Ø 3.37962	Exact station
Overall length	48.040	Z −3.700 to Z 44.340	Exact

Neck radius is $2.450 - (9.030 - Z_{P07}) \cdot \tan 30^\circ$. $Z_{P07} = 8.579337572031$ mm and the neck radius is 2.189809925899 mm. Tip radius is the neck radius minus 0.500 mm.

Circular edges

STEP	Diameter	Centre X	Centre Y	Centre Z	Axis
#27	R 8.000 meridian	0.000	0.000	4.300	(0.000, -1.000, 0.000)
#251	Ø 9.500	0.000	0.000	0.000	(0.000, 0.000, 1.000)
#286	Ø 5.500	0.000	0.000	0.000	(0.000, 0.000, 1.000)
#319	R 0.300 meridian	2.750	0.000	0.300	(0.000, 1.000, 0.000)
#341	Ø 4.900	0.000	0.000	0.300	(0.000, 0.000, 1.000)
#396	Ø 4.900	0.000	0.000	8.579338	(0.000, 0.000, 1.000)
#451	Ø 4.37962	0.000	0.000	9.030	(0.000, 0.000, 1.000)
#506	Ø 4.37962	0.000	0.000	43.840	(0.000, 0.000, 1.000)
#561	Ø 3.37962	0.000	0.000	44.340	(0.000, 0.000, 1.000)

Circles whose axis is Z are the part diameters. The two meridian circles are the sphere great-circle seam (R 8.000, lying in XZ) and the R0.300 round seam.

Every 3D coordinate in the STEP file

The file contains 107 three-dimensional CARTESIAN_POINT entities. Points within 0.00000001 mm of each other are one row (57 unique positions). 70 further points are two-dimensional (u, v) parameters of pcurves. Those are not positions on the part and are not listed.

Role "vertex" is a boundary dimension point (the P-points). "origin" is a circle, surface, or axis origin. "line point" lies on an unbounded line used as an edge. "spline control" is a B-spline control point of the base blend; except where it is also a vertex, it is not a point on the finished surface.

#	STEP ids	Role	X	Y	Z
1	#23	P01, vertex	0	0	-3.7
2	#25, #56, #69, #94, #107, #151	P02, spline control, vertex	4.578947368421	0	-2.259972636927
3	#122, #135, #148, #153, #170, #183	P03, spline control, vertex	4.75	0	-1.931974005081
4	#284	P05, vertex	2.75	0	0
5	#226	P04, vertex	4.75	0	0
6	#317, #376	P06, line point, vertex	2.45	0	0.3
7	#373, #431	P07, line point, vertex	2.45	0	8.579337572031
8	#428, #486	P08, line point, vertex	2.189809925899	0	9.03
9	#483, #541	P09, line point, vertex	2.189809925899	0	43.84
10	#538	P10, vertex	1.689809925899	0	44.34
11	#194	origin	0	0	-2.137196594792
12	#12, #253, #266, #288	origin	0	0	0
13	#301, #343, #356	origin	0	0	0.3
14	#321	origin	2.75	0	0.3
15	#29, #35	origin	0	0	4.3
16	#398, #411	origin	0	0	8.579337572031
17	#453, #466	origin	0	0	9.03
18	#508, #521	origin	0	0	43.84
19	#563, #576	origin	0	0	44.34
20	#229	line point	4.75	0	-2.137196594792
21	#63, #101	spline control	-6.840559767452	-2.069988931622	-2.259972636933
22	#62, #100	spline control	-6.840559767392	2.069988931577	-2.259972636921
23	#64, #102	spline control	-4.818051610218	-4.552472171321	-2.259972636922
24	#61, #99	spline control	-4.818051610234	4.5524721714	-2.259972636931
25	#65, #103	spline control	-2.060317672183	-6.253063381818	-2.259972636929
26	#60, #98	spline control	-2.060317672201	6.25306338175	-2.259972636925
27	#66, #104	spline control	1.131943786637	-6.008019114016	-2.259972636926
28	#59, #97	spline control	1.131943786661	6.008019114049	-2.259972636927
29	#67, #105	spline control	3.45168725538	-4.412480864862	-2.259972636927
30	#58, #96	spline control	3.451687255367	4.412480864853	-2.259972636927
31	#68, #106	spline control	4.578947368419	-2.181091624179	-2.259972636927
32	#57, #95	spline control	4.578947368423	2.181091624179	-2.259972636927
33	#115	spline control	-7.096097919698	-2.147316104364	-2.140575645422
34	#114	spline control	-7.096097919609	2.147316104313	-2.140575645414
35	#116	spline control	-4.998036296798	-4.722535786907	-2.140575645415
36	#113	spline control	-4.998036296832	4.722535786996	-2.140575645421
37	#117	spline control	-2.137283562241	-6.486654829996	-2.140575645419
38	#112	spline control	-2.137283562251	6.486654829919	-2.140575645417
39	#118	spline control	1.174229043036	-6.232456609651	-2.140575645418
40	#111	spline control	1.174229043059	6.232456609688	-2.140575645418
41	#119	spline control	3.580629595379	-4.57731492016	-2.140575645418
42	#110	spline control	3.580629595366	4.57731492015	-2.140575645418
43	#120	spline control	4.749999999999	-2.262569184852	-2.140575645418
44	#108, #121, #152	spline control	4.75	0	-2.140575645418
45	#109	spline control	4.750000000002	2.262569184853	-2.140575645418
46	#129, #177	spline control	-7.096097919691	-2.147316104366	-1.931974005084
47	#128, #176	spline control	-7.096097919617	2.147316104317	-1.931974005077
48	#130, #178	spline control	-4.998036296802	-4.722535786904	-1.931974005078

#	STEP ids	Role	X	Y	Z
49	#127, #175	spline control	-4.998036296826	4.722535786993	-1.931974005083
50	#131, #179	spline control	-2.137283562238	-6.48665483	-1.931974005082
51	#126, #174	spline control	-2.137283562254	6.486654829921	-1.93197400508
52	#132, #180	spline control	1.174229043035	-6.232456609648	-1.93197400508
53	#125, #173	spline control	1.17422904306	6.232456609688	-1.931974005081
54	#133, #181	spline control	3.580629595379	-4.577314920161	-1.931974005081
55	#124, #172	spline control	3.580629595366	4.57731492015	-1.931974005081
56	#134, #182	spline control	4.749999999998	-2.262569184851	-1.931974005081
57	#123, #171	spline control	4.750000000002	2.262569184853	-1.931974005081

Raw STEP literals are shown so driven values are not rounded away. A coordinate whose magnitude is below 1e-9 is numerical noise and is written as 0.
Model uncertainty, entity #594, is 0.00001 mm.